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PROFESSIONAL EXPERIENCE

2025 – Present, Research Associate, Woods Hole Oceanographic Institution, Woods Hole, MA, Department of Marine Chemistry and Geochemistry

2022 – 2025, Postdoctoral Investigator, Woods Hole Oceanographic Institution, Woods Hole, MA, Department of Marine Chemistry and Geochemistry

EDUCATION

University of Tennessee	Knoxville, US
Ph.D. Environmental Engineering	Aug. 2022

University of Chinese Academy of Sciences	Beijing, China
M.S. Environmental Science	Jul. 2017

University of Copenhagen	Copenhagen, Denmark
M.S. Water and Environment	Mar. 2017

University of Jinan	Jinan, China
B.Eng. Environmental Engineering	Jun. 2014

RESEARCH INTERESTS

- Microbial and chemical processes in environmental systems
- Environmental pollution and contaminant transformation
- Fate, persistence, and cycling of anthropogenic materials
- Biodegradation, bioconversion, and resource recovery
- Quantitative and molecular methods for environmental processes
- Predictive modeling across environmental systems

PUBLICATIONS

Peer-reviewed Papers:

1. Sun, Y., Wu, X., Zanina, O. G., Rivkina, E. M., Lloyd, K. G., Löffler, F. E., and Vishnivetskaya, T. A. 2026. Incomplete denitrifying bacteria drive N₂O fluxes in ancient Siberian permafrost microcosms. *FEMS Microbiology Ecology*. 102(5), fiag034. <https://doi.org/10.1093/femsec/fiag034>
2. Liu, S., Li, Y., Zeng, X., Sun, Y., Huang, Z., Du, C., Lang, J., Wang, S., and Jia, Y. 2026. Reversible control of microbial As (III) oxidation by nitrous oxide availability in flooded paddy soils. *Environmental Science & Technology*. In press. <https://doi.org/10.1021/acs.est.6c04988>
3. Zhou, J., Sang, C., Liu, Z., Li, W., Sun, Y., Zhang, J., Wang, S., and Zeng, X. 2026. Bioconversion of selenite to selenium nanoparticles by *Paenibacillus polymyxa* P10: metabolic basis and potential

for wastewater treatment. *Applied and Environmental Microbiology*. e00704-26.

<https://doi.org/10.1128/aem.00704-26>

4. Sun, Y., James, B. D., Houston, R. K., Edwards, B., Izallalen, M., Mazumde, S., Shankar R., Reddy, C. M., and Ward, C. P. 2025. Temperature controls the lifetimes and microbial communities degrading cellulose diacetate in the coastal ocean. *Environmental Science & Technology*. 59, 36, 19468–19478. <https://doi.org/10.1021/acs.est.5c05334>
5. James, B. D., Sun, Y., Pate, K., Irving, B., Ward, C. P. 2025. Strategies for designing circular, sustainable, and nonpersistent consumer plastic products: A case study of drinking straws. *Environmental Science & Technology*. 59, 34, 18177–18189. <https://doi.org/10.1021/acs.est.5c05448>
6. Murdoch, F. K., Sun, Y., Fuller, M. E., Lilly, J., Wilson, J., Mullins, L., Hill, A., Löffler, F. E., and Kucharzyk, H. K. 2025. Presence of micro and nanoplastics affects degradation of chlorinated solvents. *Toxics*, 13(8), 656. <https://doi.org/10.3390/toxics13080656>
7. Waldrop, M. P., Ernakovich, J. G., Schaefer, S. R., Mackelprang, R., Tas, N., Vishnivetskaya, T. A., Li, Z., Ricketts, M. P., Leewis, M. C., Hewitt, R., Hultman, J., Sun, Y., et al. 2025. Microbial ecology of permafrost soils: populations, processes, and perspectives. *Permafrost and Periglacial Processes*, 0:1–14. <https://doi.org/10.1002/ppp.2264>
8. Sun, Y., Yin, Y., He, G., Cha, G., Ayala-del-Río, H., Konstantinidis, K. T., and Löffler, F. E. 2024. pH selects for distinct N₂O-reducing microbiomes in tropical soil microcosms. *ISME Communications*, ycae070. <https://doi.org/10.1093/ismeco/ycae070>
9. James, B. D., Sun, Y., Pate, K., Perri, S. T., Shankar, M., Reddy, C. M., and Ward, C. P. 2024. Foaming enables material-efficient bioplastic products with minimal persistence. *ACS Sustainable Chemistry & Engineering*, 12, 43, 16030–16040. <https://doi.org/10.1021/acssuschemeng.4c05822>
10. Sun, H., Radosevich, M., Sun, Y., Millet, L., Qian, S., and Zhuang, J. 2024. Pore confinement enhances but surface adhesion reduces bacterial cell-to-cell interaction. *Biology and Fertility of Soils*, 1-10. <https://doi.org/10.1007/s00374-024-01841-w>
11. Yin, Y., Murdoch F. K., Murdoch R., Yan J., Chen G., Xie Y., Sun Y., and Löffler, F. E. 2024. Nitrous oxide inhibition of methanogenesis represents an underappreciated greenhouse gas emission feedback. *The ISME Journal*, p.wrae027. <https://doi.org/10.1093/ismejo/wrae027>
12. James, B. D., Sun, Y., Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2024. Strategies to reduce the environmental lifetimes of drinking straws in the coastal ocean. *ACS Sustainable Chemistry & Engineering*, 12, 6, 2404–2411. <https://doi.org/10.1021/acssuschemeng.3c07391>
13. Yang, S., Shobnam, N., Sun, Y., Löffler, F. E., and Im, J. 2024. The relative contributions of Mn(III) and Mn(IV) in manganese dioxide polymorphs to bisphenol A degradation. *Journal of Hazardous Materials*, 461, 132596. <https://doi.org/10.1016/j.jhazmat.2023.132596>
14. Sun, Y., Mazzotta, M. G., Miller, C. A., Apprill, A., Izallalen, M., Mazumde, S., Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2023. Distinct microbial communities degrade cellulose diacetate bioplastics in the coastal ocean. *Applied and Environmental Microbiology*, e01651-23. <https://doi.org/10.1128/aem.01651-23>
15. Zhang, L., Yin, Y., Sun, Y., Liang, X., Graham, D., Pierce, E; Löffler, F. E., and Gu, B. 2023. Inhibition of methylmercury and methane formation by nitrous oxide in Arctic tundra soil microcosms. *Environmental Science & Technology*, 57(14), 5655-5665. <https://doi.org/10.1021/acs.est.2c09457>

16. Sun, Y., Wang, C., May, A. L., Chen, G., Yin, Y., Im, J., Campagna, S., and Löffler, F. E. 2023. Mn(III)-mediated bisphenol A degradation: Mechanisms and products. *Water Research*, 235, 119787. <https://doi.org/10.1016/j.watres.2023.119787>
17. Xie, Y., Ramirez, D., Chen, G., He, G., Sun, Y., Murdoch, F. K., and Löffler, F. E. 2023. Genome-wide expression analysis unravels fluoroalkane metabolism in *Pseudomonas* sp. strain 273. *Environmental Science & Technology*, 57(42), 15925–15935. <https://doi.org/10.1021/acs.est.3c03855>
18. Sun, H., Sun, Y., Jin, M., Ripp, S. A., Saylor, G. S., and Zhuang, J. 2022. Domestic plant food loss and waste in the United States: Environmental footprints and mitigation strategies. *Waste Management*, 150, 202-207. <https://doi.org/10.1016/j.wasman.2022.07.006>
19. Sun, Y., Im, J., Shobnam, N., Fanourakis, S. K., He, L., Anovitz, L. M., Erickson R. P., Sun, H., Zhuang, J., and Löffler, F. E. 2021. Degradation of adsorbed bisphenol A by soluble Mn(III). *Environmental Science & Technology*, 55(19), 13014-13023. <https://doi.org/10.1021/acs.est.1c03862>
20. Shobnam, N., Sun, Y., Mahmood, M., Löffler, F. E., and Im, J. 2021. Biologically mediated abiotic degradation (BMAD) of bisphenol A by manganese-oxidizing bacteria. *Journal of Hazardous Materials*, 417, 125987. <https://doi.org/10.1016/j.jhazmat.2021.125987>
21. Sun, Y., Zeng, X., Yang, L., Shi, Y., Chen, X., and Zhuang, J. 2017. Effects of strong reductive process on transformation of heavy metals in protected vegetable soil. *The Journal of Applied Ecology*, 28(11), 3759-3766. <https://doi.org/10.13287/j.1001-9332.201711.032>

Manuscript in Review/in Preparation:

22. Sun, Y., Grim, S. L., James, B. D., Reddy, C. M., and Ward, C. P. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. [ChemRxiv](#). In review at *Environmental Science: Processes & Impacts*.
23. Ward, C. P., Schwartz, G., Sun, Y., Messenger, S., Frye-Jones, J., Italiane, D., Swarr, G., Aguilera, M., Rodgers, R. Chronic sunlight-driven mobilization of persistent, bioactive carbon and vanadium from asphalt to downstream aquatic systems. In review at *PNAS*.
24. Sun, Y., Ahern, O., Huber, J., Reddy, C. M., and Ward, C. P. Stable-isotope probing of microbial species driving cellulose diacetate biodegradation in the coastal ocean. In preparation.
25. Sun, Y., Reddy, C. M., and Ward, C. P. A next-generation method for quantifying plastic biodegradation in seawater. In preparation.
26. Sun, Y., Reddy, C. M., and Ward, C. P. Boosting biodegradation of cellulose diacetate bioplastic by graphitic carbon nitride and sunlight in seawater. In preparation.

Conference Papers:

1. Sun, Y., Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Is adsorbed bisphenol A (BPA) degradable? *Eleventh International Conference on the Remediation and Management of Contaminated Sediments*. Best paper award.
2. Sun, Y., Im, J., He, L., Erickson R. P., and Löffler, F. E. 2022. Soluble Mn(III)-mediates degradation of adsorbed bisphenol A. *Thirteenth Annual Geosyntec Student Paper Competition*. Third-place award.

RESEARCH FUNDING AND GRANT EXPERIENCE

Pending Proposals:

1. Collaborative research: Assessing the impacts of plastic-derived organic carbon on aquatic microbiomes across the freshwater–ocean continuum. 2026. **PI.** NSF-CBET. \$491,358.
2. Fueling a circular plastics economy: AI-guided conversion of bioplastic waste to bioelectricity. 2026. **PI.** ACS Early Career Postdoctoral–Faculty Bridge Grant. \$125,000.
3. Ocean-derived esterases for cellulose diacetate waste treatment and biodegradable material development. 2026. **PI.** WHOI Innovative Technology Program. \$99,803.
4. Analysis of microbial communities in enrichment experiments with cellulose esters. 2026. **Co-PI.** Eastman Chemical Company. \$45,000.

Awarded Funding:

5. Characterizing the Distribution of Bisphenol A in Pores of Microscale Activated Carbon. Neutron Sciences General User Proposal (IPTS #24752.1), Oak Ridge National Laboratory, 2020. **PI.**

HONORS AND AWARDS

- American Society for Microbiology Future Leaders Mentorship Fellowship, 2024
- Extraordinary Professional Promise Award, University of Tennessee, 2022
- Paper Competition Award, Geosyntec Consultants, Inc., 2022
- Paper Competition Award, Battelle Sediments Conference, 2022
- Graduate Student Senate Academic Support, University of Tennessee, 2022
- Graduate Student Senate Travel Award, University of Tennessee, 2019-2022
- Merit student, University of Chinese Academy of Sciences, 2016
- Outstanding Student Scholarship, University of Jinan, 2010-2013

PARTICIPATION IN EDUCATION PROGRAM

- ASM Future Leaders Mentorship Fellowship Program, 2024-2026
- WHOI Summer Student Fellowship Program, May-Aug 2024, 2025
- Geosciences Education & Mentorship Support Program, 2023-2024
- *Inclusive Teaching Certificate*, University of Tennessee, 2023
- *Course-Based Assessment Certificate*, University of Tennessee, 2023
- Student Mentoring and Research Training Program, University of Tennessee–Oak Ridge Innovation Institute, May-Aug 2022
- Summer School for High School Students, China, Jul-Aug 2014

MENTORING

- Kyle DeCruccio, undergraduate student, Cape Cod Community College, Jun-Aug, 2025
- Mallory Kastner, Ph.D. student, MIT-WHOI Joint Program, Feb-Mar, 2025
- Dominic Italiane, undergraduate student, University of Massachusetts-Boston, 2024-2025
- Shinyeong Choe, Ph.D. student, Woods Hole Oceanographic Institution, Feb-Mar, 2024
- Kimberly Lutchman, undergraduate student, Queens College, CUNY, Aug 2023-2024
- Devan Linson, undergraduate student, Washington State University, May-Aug, 2022

- Divinity Oliver, undergraduate student, Washington State University, May-Aug 2022
- Jessica Rattray, undergraduate student, Vassar College, May-Aug, 2022

PROFESSIONAL DEVELOPMENT AND SERVICES

Professional Development:

- ACS Postdoc to Faculty Workshop, 2026
- Proposal Writing Workshop, Woods Hole Oceanographic Institution, 2023
- Academic Job Application Workshop, Woods Hole Oceanographic Institution, 2023

Service:

- AEESP Experts Program Committee, 2026–present.
- Poster judge, 2025 AEESP Conference.
- Reviewer for *ACS Environmental Au*, *Bioresource Technology*, *Chemosphere*, *Communications Earth & Environment*, *Ecotoxicology*, *Environmental Sciences Europe*, *Environmental Geochemistry and Health*, *Environmental Monitoring and Assessment*, *Environmental Research*, *Environmental Science & Technology*, *Frontiers in Microbiology*, *Frontiers in Marine Science*, *Journal of Geophysical Research – Biogeosciences*, *Journal of Hazardous Materials Advances*, *Journal of Hazardous Materials Letters*, *Reviews of Environmental Contamination and Toxicology*, and *Water Research*.

Outreach:

- Massachusetts Region V Science and Engineering Fair, 2024
- Reverse Science Fair, Falmouth High School, MA, 2023

Professional Affiliation:

- American Chemical Society
- American Geophysical Union
- American Society for Microbiology
- Association of Environmental Engineering and Science Professors

SELECTED PRESENTATIONS (presenter underlined)

Oral:

1. Sun, Y., Ahern, O., Huber, J., Reddy, C. M., and Ward, C. P. 2026. Stable-isotope probing of microbial species driving cellulose diacetate biodegradation in the coastal ocean. American Chemical Society Spring Meeting, Atlanta, GA.
2. Sun, Y. Environmental pollution and health: insights from plastic degradation. 2025. Morgan State University, Baltimore, MD. *Invited*.
3. Sun, Y. Unravelling plastic degradation in the coastal ocean. 2025. Woods Hole Oceanographic Institution, Woods Hole, MA. *Invited*.
4. Sun, Y., Frye-Jones, J., Grim, S., James, B., Reddy, C. M., and Ward, C. P. 2025. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. AEESP Conference, Durham, NC.

5. **Sun, Y.**, James, B., Reddy, C. M., and Ward, C. P. 2025. Temperature controls the timelines and microbial communities degrading cellulose diacetate in the coastal ocean. American Chemical Society Spring Meeting, San Diego, CA.
6. **Sun, Y.**, Grim, S., James, B., Reddy, C. M., and Ward, C. P. 2025. Photodegradation of plastics shapes the diversity of microbial communities and functions in seawater. American Chemical Society Spring Meeting, San Diego, CA.
7. **Sun, Y.** Potential of microbial degradation of plastics in the ocean. 2025. AAAS Annual Meeting, Boston, MA.
8. **Sun, Y.** Understanding the fate of conventional and next-generation plastics in the ocean. 2024. University of Massachusetts, Dartmouth, MA. *Invited.*
9. **Sun, Y.**, Mazzotta, M. G., Miller, C. A., Apprill, A., Izallalen, M., Mazumde, S., Perri, S. T., Edwards, B., Reddy, C. M., and Ward, C. P. 2024. Unique microbes degrade cellulose diacetate bioplastics in the ocean. American Chemical Society Spring Meeting, New Orleans, LA.
10. **Sun, Y.**, Ward, C. P., James, B., and Reddy, C. M. 2023. Translating fundamental knowledge about the degradation of cellulose diacetate in the environment into high-utility, low-persistence consumer products. Eastman Chemical Company, Kingsport, TN. *Invited.*
11. **Sun, Y.** 2023. “Bioplastics meet biogeochemical cycles”, WHOI's Microplastics Initiative. Woods Hole Oceanographic Institution, Woods Hole, MA.
12. **Sun, Y.**, Mazzotta, M. G., Miller, C. A., Apprill, A., Reddy, C. M., and Ward, C. P. 2023. Distinct microbial communities degrade cellulose diacetate plastics in the coastal ocean. AEESP Conference, Boston, MA.
13. **Sun, Y.** 2022. “Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A”, webinar presentation. Geosyntec Consultants, Inc. *Invited.*
14. **Sun, Y.**, Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A. Eleventh International Conference on the Remediation and Management of Contaminated Sediments, Nashville, TN. *Invited.*

Poster:

15. **Sun, Y.**, James, B., Ahern, O., Huber, J., Reddy, C. M. and Ward, C. P. 2026. A systematic evaluation of consumer plastics degradation in coastal environments. Gordon Research Conference, Environmental Sciences: Water. Holderness, NH.
16. **Sun, Y.**, Wu, X., Zanina, O., Rivkina, E., Lloyd, K., Löffler, F. E., and Vishnivetskaya, T. 2022. Nitrous oxide emissions from thawing permafrost: A microcosm study. American Geophysical Union Fall Meeting, Chicago, IL.
17. **Sun, Y.**, Yin, Y., Chen, G., Cha, G., Ayala-del-Río, H., Konstantinidis, K. T., and Löffler, F. E. 2022. pH selects for distinct N₂O-reducing microbiomes in tropical soil microcosms. Eighteenth International Symposium on Microbial Ecology, Lausanne, Switzerland.
18. **Sun, Y.**, Im, J., He, L., Erickson, P., and Löffler, F. E. 2022. Role of dissolved Mn(III) in the transformation of adsorbed bisphenol A. Eleventh International Conference on the Remediation and Management of Contaminated Sediments, Nashville, TN.
19. **Sun, Y.**, Shobnam, N., Im, J., Chen, G., Yin, Y., Erickson, P. R., and Löffler, F. E. 2019. Degradation of adsorbed BPA by oxidized manganese species. American Geophysical Union Fall Meeting, San Francisco, CA.

20. **Sun, Y.**, Shobnam, N., Im, J., Chen, G., Yin, Y., Campagna, S., and Löffler, F. E. 2019. Biologically mediated abiotic degradation of bisphenol A. American Chemical Society Spring Meeting, Orlando, FL.